



PC1 · TRANSLATIONAL RESEARCH IN CP · SESSION 2: THE PROMISE OF BIOMARKERS

Machine Learning for CP Diagnosis: Promise and Pitfalls

Computational approaches to complex data, and what has to be true before a methylation test reaches a child

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Employee	University of Delaware
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Investigational status: every methylation classifier discussed in this talk is investigational. None is FDA-cleared, and none is available for clinical use. No off-label use of any approved product is discussed.



WHY METHYLATION

DNA methylation keeps a record of early-life stress

1 Perinatal stress

Hypoxia, infection, inflammation and growth restriction are all linked to CP.

2 Marks imprinted

The same stresses change methyl marks at CpG sites across the genome.

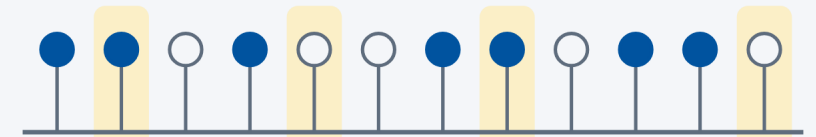
3 Marks persist

Methylation is copied at each cell division, so the change is carried forward.

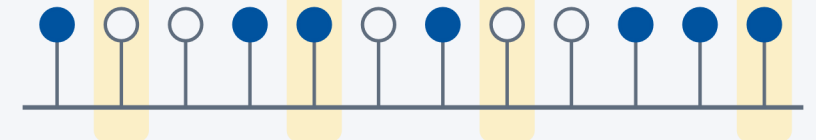
4 Readable later

Blood or muscle sampled years later can still carry the pattern.

Control



Spastic CP



Same DNA sequence · filled = methylated CpG · shaded = sites that differ

THE HONEST TEST

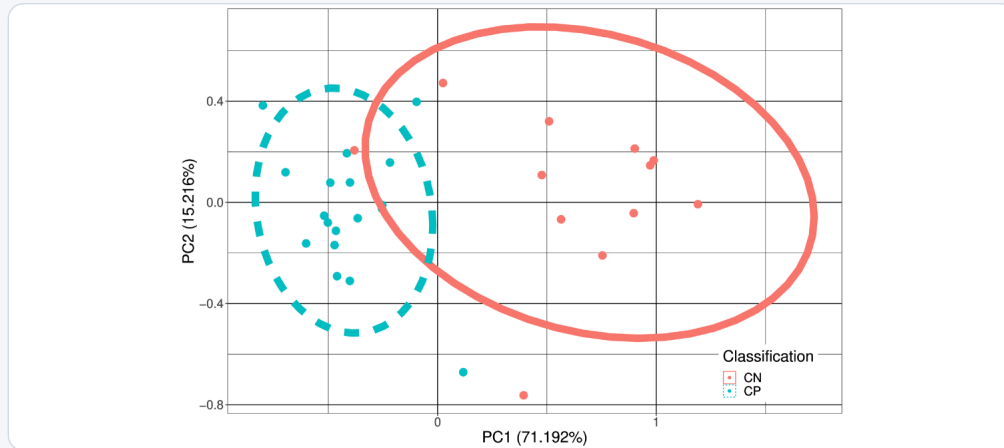
Six sites, 96 new children, a different tissue

Discovery · blood

30 children

1.00

median F1

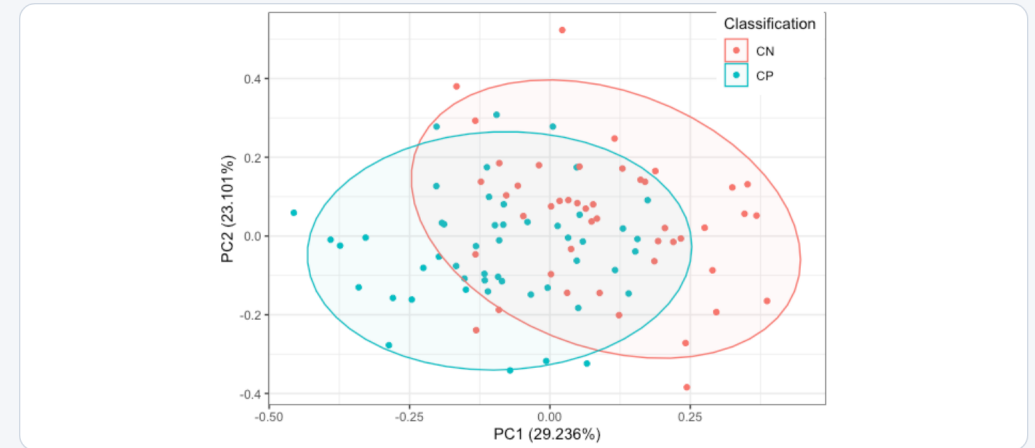


New cohort · skeletal muscle

49 CP, 47 controls

0.82

median F1



The change in median F1 from **1.00 to 0.82** is what honest, real-world validation looks like. The question now is understanding the 0.18 gap between them, and whether we can bridge it in any way.

F1 blends how often a positive call is correct with how many true cases are caught. 1.00 is perfect; 0.50 is close to guessing.

Three studies, three tissues, one recurring answer

Crowgey *et al.* 2018

BLOOD · ADOLESCENTS

- 16 with spastic CP vs 16 controls
- 6,588 of 1.47 M CpG sites differed
- Blinded 4-year-olds: 73% accuracy, AUC 0.69

Hicks *et al.* in process

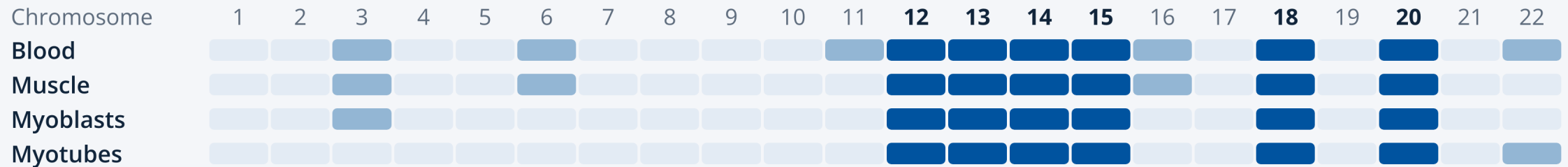
BLOOD · CP WITH EPILEPSY

- 30 children, EPIC array
- A 6-CpG panel, sites re-picked inside every split
- SVM median F1 = 1.00; epilepsy did not mask CP

Robinson *et al.* 2022

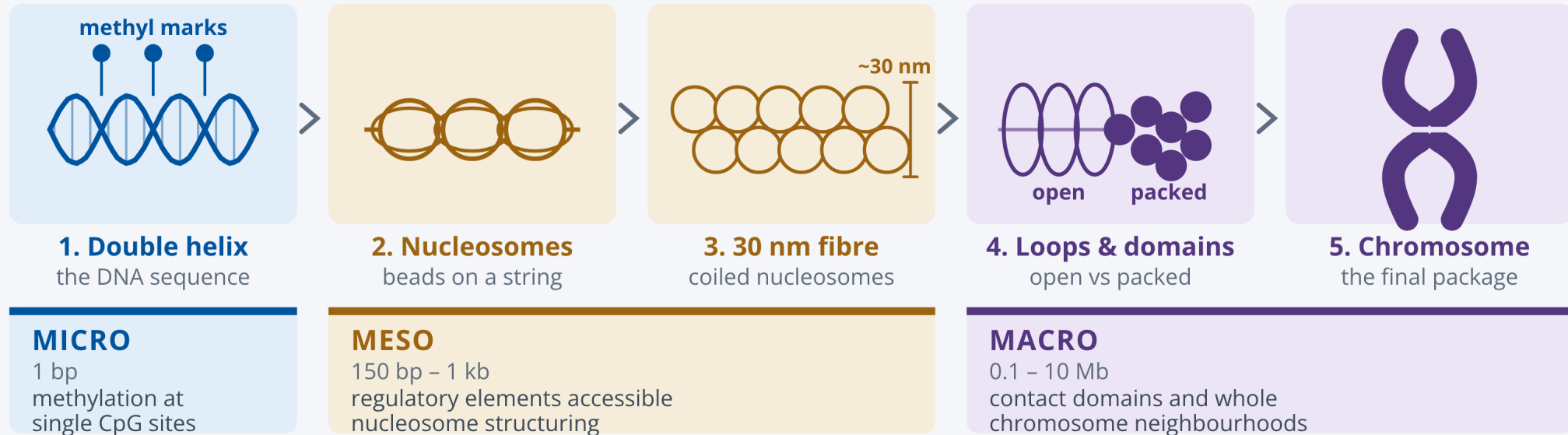
MUSCLE STEM CELLS

- 6 vs 6 surgical biopsies
- 525 sites in myoblasts, 1,774 in myotubes
- Differences persist in culture → intrinsic to the cells



Six chromosomes — **12, 13, 14, 15, 18 and 20** — are enriched in all four sample types.

One genome, packaged at four scales



The chromosome-level clustering of results on the previous slide is a **macro-scale** result — but every measurement behind it was made at the **micro** scale. **This is the missing gap.**

THE CHALLENGE

The same measurement, two different meanings

1,468,477

CpG sites per child

32

children

6,588

differentially methylated sites

At this ratio a pattern-finder will always find a pattern.

But the harder problem is not how many features there are — it is what any one of them means.

DRIVER

The gene was switched on

Domain open · promoter accessible → silenced

Something happened here.

Methylation arrives at an active promoter and silences it — a real regulatory lesion.

PASSENGER

The gene was already off

Domain closed · already repressed → still silenced

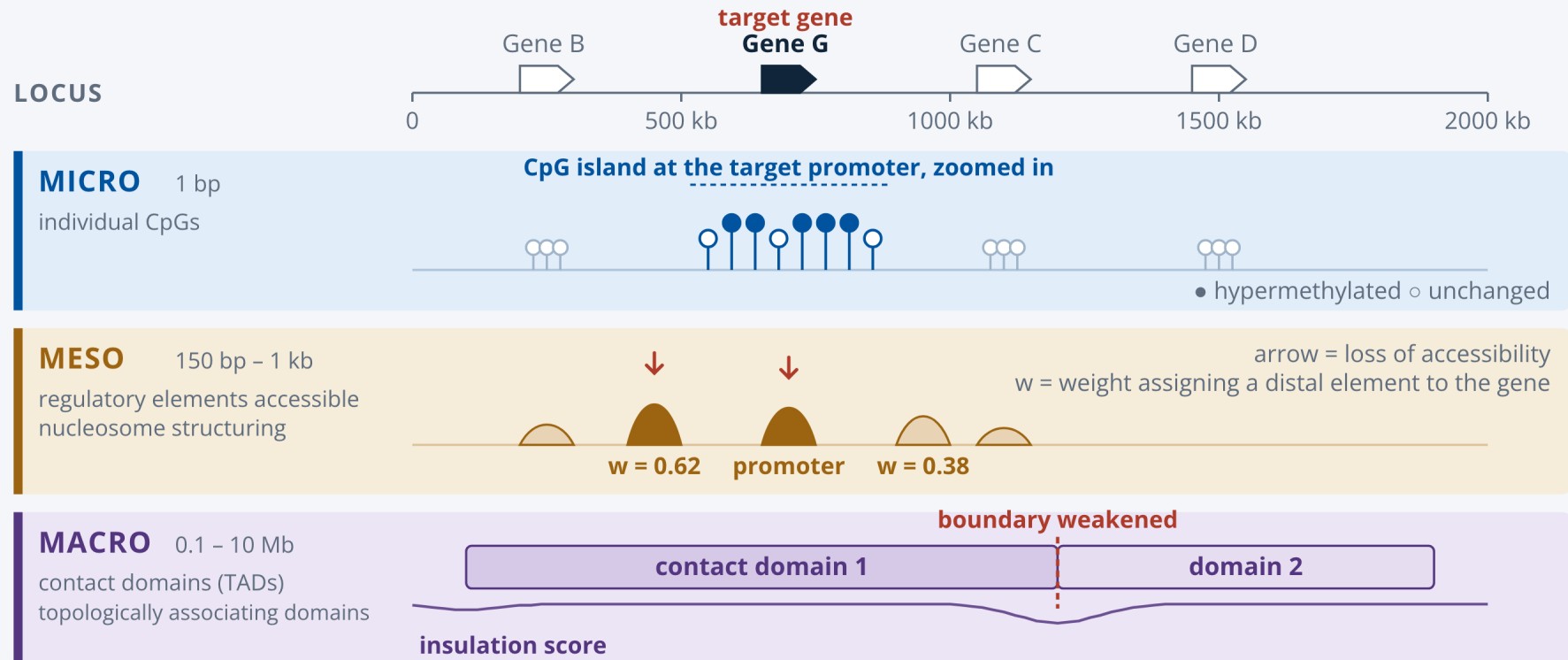
Nothing happened here.

The promoter sat in a repressed domain; methylation only locks in what the lineage had already set.

A methylation assay reports both of these identically — “promoter hypermethylated.” **A classifier trained on that one channel learns both, and reports either with the same confidence.**

WHY ONE CHANNEL IS NOT ENOUGH

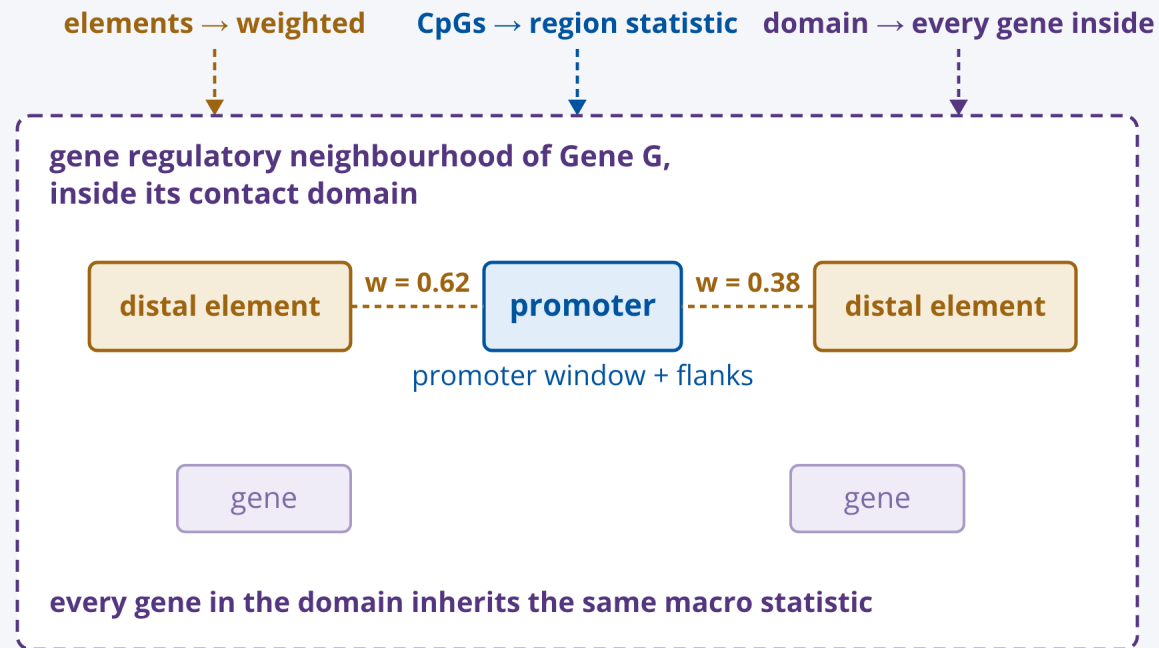
Three scales, three native resolutions, one stretch of genome



A real lesion marks **all three** channels. An artefact of cell lineage does not.

THE METHOD

Reduce three channels to one score per gene — then test as usual



Marsh & Doane, *Curr Issues Mol Biol* 2026;48:915

Measure the correlation, don't assume it

The three assays share one specimen, so they share nuisance variation. Their null correlation is estimated by shuffling sample labels — not assumed to be zero.

	micro	meso	macro
micro	1	0.58	0.35
meso	0.58	1	0.47
macro	0.35	0.47	1

Illustrative values.

One calibrated score per gene goes into ordinary pathway testing. A result now means **these genes change together across scales** — not that many changed somehow.

PITFALLS

Five questions to ask of any AI-driven diagnostic

Pitfall	What it looks like	Ask
Too few patients, too many features	A perfect score from 30 children and a million sites	How many patients? Was it replicated?
Information leak	Sites chosen using all samples, then “tested” on some of them	Was selection done inside the training data only?
One channel only	A driver lesion and a lineage artefact scored identically	Is there corroborating evidence of another kind?
Wrong population	Built in adolescents, needed in infants	Tested at the age and in the setting of use?
Flattering metric	73% accuracy hid 40% specificity	Sensitivity, specificity and PPV all reported?

READING A RESULT

Prevalence decides what a positive result means

An illustrative test with 95% sensitivity and 95% specificity — better than anything shown today, and not our assay:

High-risk follow-up clinic

Illustrative 1 in 10 · per 1,000 infants: 100 with CP

95

45

true positives false positives

Chance a positive is truly CP: **≈ 68%**

The same test becomes clinically useful.

All newborns

CP ≈ 1 in 345 · per 10,000 babies: 29 with CP

28

499

true positives false positives

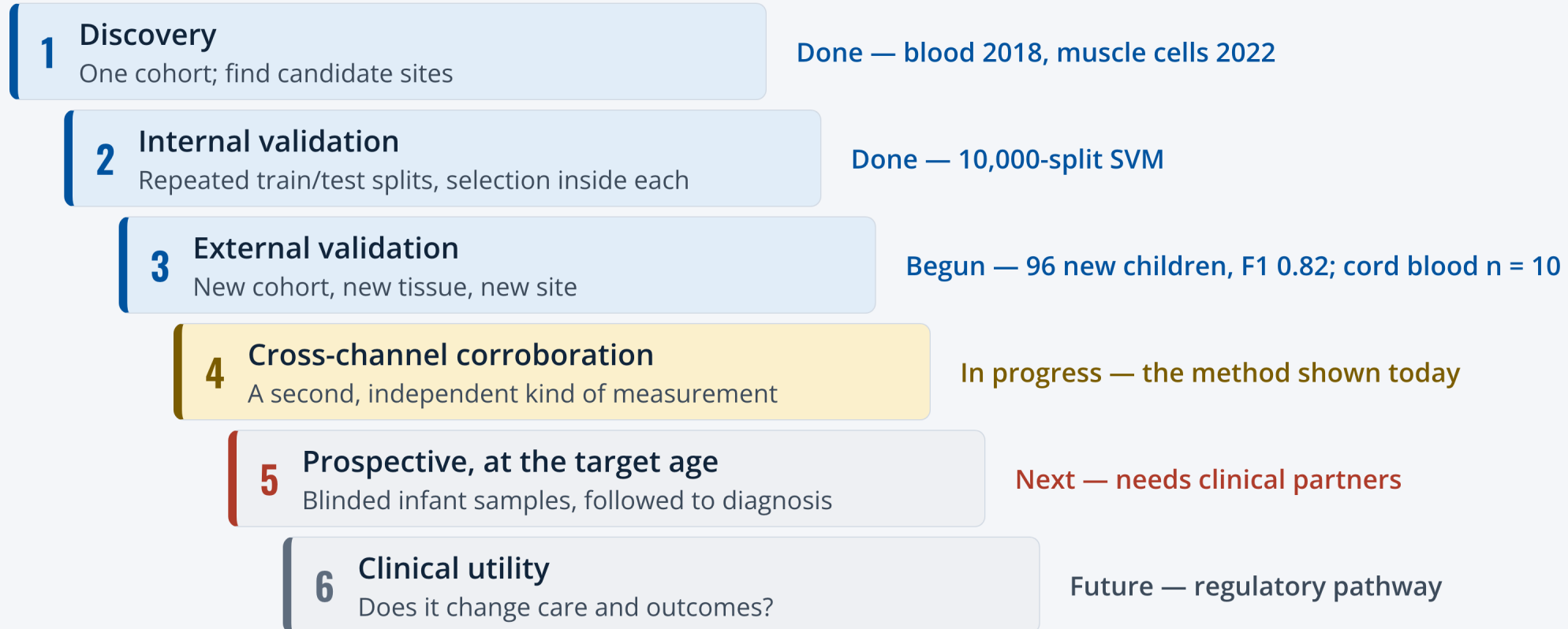
Chance a positive is truly CP: **≈ 5%**

About 19 of every 20 positives are false alarms.

The first clinical use of a methylation test is going to be in an **at-risk infant population**, alongside the General Movements Assessment and imaging.

VALIDATION REQUIREMENTS

From a promising panel to a clinical test



TAKE-HOME

Three things to carry into the panel

- 1 The signal is real.** Blood and muscle methylation differ in spastic CP, and a six-site panel held up in an independent cohort of 96 children.
- 2 One measurement is ambiguous.** A single channel cannot separate a real lesion from a record of cell lineage. Corroboration across scales can, and it is measurable rather than assumed.
- 3 Clinicians make the next step possible.** Consented samples in early infancy, followed to diagnosis, are what this program now needs more than any new algorithm.

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Thank you

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Bioinformatics & Computational Biology

Key papers

Crowgey *et al.* 2018, *BMC Bioinformatics* 19:225

Robinson *et al.* 2022, *J Pers Med* 12:1978

Hicks *et al.*, CP with epilepsy methylation classifier [journal, year]

Marsh & Doane 2026, *Curr Issues Mol Biol* 48:915

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